

SL8521E Uboot LCD MIPI 驱动的 Display 寄存器介绍

文档版本

V1.1

发布日期

2020-07-15

版权所有 © 紫光展锐科技有限公司。保留一切权利。

本文件所含数据和信息都属于紫光展锐所有的机密信息，紫光展锐保留所有相关权利。本文件仅为信息参考之目的提供，不包含任何明示或默示的知识产权许可，也不表示有任何明示或默示的保证，包括但不限于满足任何特殊目的、不侵权或性能。当您接受这份文件时，即表示您同意本文件中内容和信息属于紫光展锐机密信息，且同意在未获得紫光展锐书面同意前，不使用或复制本文件的整体或部分，也不向任何其他方披露本文件内容。紫光展锐有权在未经事先通知的情况下，在任何时候对本文件做任何修改。紫光展锐对本文件所含数据和信息不做任何保证，在任何情况下，紫光展锐均不负责任任何与本文件相关的直接或间接的、任何伤害或损失。

请参照交付物中说明文档对紫光展锐交付物进行使用，任何人对紫光展锐交付物的修改、定制化或违反说明文档的指引对紫光展锐交付物进行使用造成的任何损失由其自行承担。紫光展锐交付物中的性能指标、测试结果和参数等，均为在紫光展锐内部研发和测试系统中获得的，仅供参考，若任何人需要对交付物进行商用或量产，需要结合自身的软硬件测试环境进行全面的测试和调试。非经本公司书面许可，任何单位和个人不得擅自摘抄、复制本文档内容的部分或全部，并不得以任何形式传播。

紫光展锐科技有限公司

前言

概述

本文档描述了 S18521E uboot mipi lcd 驱动所涉及到的 DSI 和 DPU 相关寄存器使用说明。

读者对象

本文档主要适用于手机厂商 Display/LCM 驱动开发者，为他们提供 DPU/DSI 方面的技术支持。

缩略语

缩略语	英文全名	中文解释
DSI	Display Serial Interface	显示串行接口
DPU	Display Process Unit	显示控制单元

目 录

目录

前 言	ii
目 录	iii
1 DPU 寄存器	4
1.1 mipi info struct	4
1.2 mipi timing struct	5
1.3 panel spec struct	7
2 DSI 寄存器	9
2.1 Init()函数	9
2.2 mipi info	14
2.3 mipi_timing	19
2.4 panel spec struct	22

1 DPU 寄存器

DPU 寄存器基地址为 0x20800000，可以直接写值。

以 /bsp/u-boot15_sprdroidq/drivers/video/sprdfb/lcd/ lcd_st7782m_mipi.c 驱动为例。

1.1 mipi info struct

```
static struct info_mipi lcd_st7782m_mipi_info = {
    .work_mode = SPRDFB_MIPI_MODE_VIDEO,
    .video_bus_width = 24, /*18,16*/
    .lan_number = 1,
    .phy_feq = 300*1000,
    .h_sync_pol = SPRDFB_POLARITY_POS,
    .v_sync_pol = SPRDFB_POLARITY_POS,
    .de_pol = SPRDFB_POLARITY_POS,
    .te_pol = SPRDFB_POLARITY_POS,
    .color_mode_pol = SPRDFB_POLARITY_NEG,
    .shut_down_pol = SPRDFB_POLARITY_NEG,
    .timing = &lcd_st7782m_mipi_timing,
    .ops = NULL,
};
```

相关参数：

work_mode

video_bus_width

h_sync_pol

v_sync_pol

de_pol te_pol

寄存器：

```
#define DISPC_DPI_CTRL (0x0170)
```

1.1.1.79 DPI_CTRL

0x00000170			DPI Config (0x00000000)									DPI_CTRL				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	reserved															DPI_HALT_EN
Type	RO															RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DSI_DPI_COLO_RM	DSI_DPI_SHDW	reserved			EDPI_TE_SEL	EDPI_TE_POL	EDPI_TE_EN	DPI_BITS		reserved					
Type	RW	RW	RO			RW	RW	RW	RW		RO					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DPI_CTRL

Field Name	Bit	Type	Reset Value	Description
reserved	[31:17]	RO	0x0	reserved
DPI HALT EN	[16]	RW	0x0	DPI halt function enable.
DSI DPI COLO RM	[15]	RW	0x0	DSI DPI I/F color mode 0: Normal color mode;
DSI DPI SHDW	[14]	RW	0x0	1: Reduced colormode. DSI DPI I/F shut down 0: No action; 1: DPI is shut down.
reserved	[13:11]	RO	0x0	reserved
EDPI TE SEL	[10]	RW	0x0	eDPI TE selection 0: From eDPI I/F; 1: From external PAD.
EDPI TE POL	[9]	RW	0x0	eDPI TE polarity 0: Positive pulse; 1: Negative pulse.
EDPI TE EN	[8]	RW	0x0	eDPI TE enable 0: Disable; 1: Enable.
DPI BITS	[7:6]	RW	0x0	DPU pixel data width 0: 16bits; 1: 18bits; 2: 24bits; 3: reserved.
reserved	[5:0]	RO	0x0	reserved

1.2 mipi timing struct

```
static struct timing_rgb lcd_st7782m_mipi_timing = {
    .hfp = 80, /* unit: pixel */
    .hbp = 80,
    .hsync = 20,
    .vfp = 4, /*unit: line*/
}
```

```
.vbp = 4,
.vsync = 1,
};
```

相关参数:

hfp :[31:20]

hbp :[19:8]

hsync :[7:0]

寄存器:

```
#define DISPC_DPI_H_TIMING (0x0174)
```

1.1.1.80 DPI_H_TIMING

0x00000174			DPI Horizontal Timing (0x00000000)										DPI H_TIMING			
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	HFP												HBP			
Type	RW												RW			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	HBP								HSYNC							
Type	RW								RW							
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DPI_H_TIMING

Field Name	Bit	Type	Reset Value	Description
HFP	[31:20]	RW	0x0	Horizontal front porch, pixel unit.
HBP	[19:8]	RW	0x0	Horizontal back porch, pixel unit.
HSYNC	[7:0]	RW	0x0	HSYNC width, pixel unit.

相关参数:

vfp :[31:20]

vbp :[19:8]

vsync :[7:0]

寄存器:

```
#define DISPC_DPI_V_TIMING (0x0178)
```

1.1.1.81 DPI_V_TIMING

0x00000178			DPI Vertical Timing (0x00000000)									DPI V TIMING				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	VFP											VBP				
Type	RW											RW				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	VBP							VSYNC								
Type	RW							RW								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DPI_V_TIMING

Field Name	Bit	Type	Reset Value	Description
VFP	[31:20]	RW	0x0	Vertical front porch, line unit.
VBP	[19:8]	RW	0x0	Vertical back porch, line unit.
VSYNC	[7:0]	RW	0x0	VSYNC width, line unit.

1.3 panel spec struct

```
struct panel_spec lcd_st7782m_mipi_spec = {
    .width = 240,
    .height = 320,
    .fps = 60,
    .type = LCD_MODE_DSI,
    .direction = LCD_DIRECT_NORMAL,
    .info = {
        .mipi = &lcd_st7782m_mipi_info
    },
    .ops = &lcd_st7782m_mipi_operations,
};
```

相关参数:

width:[0:11]

height:[16:27]

寄存器:

```
#define DISPC_SIZE_XY (0x0008)
```


1.1.1.3 PANEL_SIZE

0x00000008				Panel Size (0x00000000)								PANEL_SIZE				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	reserved				PANEL_SIZE_Y											
Type	RO				RW											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	reserved				PANEL_SIZE_X											
Type	RO				RW											
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PANEL_SIZE

Field Name	Bit	Type	Reset Value	Description
reserved	[31:28]	RO	0x0	reserved
PANEL_SIZE_Y	[27:16]	RW	0x0	Panel vertical size. It is pixel unit.
reserved	[15:12]	RO	0x0	reserved
PANEL_SIZE_X	[11:0]	RW	0x0	Panel horizontal size. It is pixel unit.

width 和 height 还有 type 参数还涉及到 dpi clock 计算写入:

源文件: source/bsp/u-boot15_sprdroidq/drivers/video/sprdfb/sprdfb_dispc.c

```
static void dispc_update_clock(struct sprdfb_device *dev){
```

```
...
```

```
    dispc_dpi_clk_set(DISPC_DPI_SEL_DEFAULT, (dividor-1));
```

```
...
```

```
}
```

/arch/arm/include/asm/arch-sharkle/chip_sharkle/ap_clk_core.h

```
#define REG_AP_CLK_CORE_CGM_DISPC0_CFG ( CTL_BASE_AP_CLK_CORE +  
0x00A0 )
```

相关参数: fps

在/bsp/u-boot15_sprdroidq/drivers/video/sprdfb/sprdfb_dispc.c 以下函数用于 DPI CLOCK 计算:

```
static void dispc_update_clock(struct sprdfb_device *dev)
```

```
{
```

```
...
```

```
    need_clock = hpixels * vlines * panel->fps;
```

```
    divisor = SPRDFB_DPI_CLOCK_SRC/need_clock;
```

```
    if(SPRDFB_DPI_CLOCK_SRC - divisor*need_clock > (need_clock/2) ) {
```

```
        divisor += 1;
```

```
    }
```

...

}

相关参数: direction

暂时没有用到。

2 DSI 寄存器

DSI 寄存器基地址: 0x21800000

参考文档: SL8521E DSI 控制器介绍 1.0.pdf

以 /bsp/u-boot15_sprddroidq/drivers/video/sprdfb/lcd/ lcd_st7782m_mipi.c 驱动为例。

2.1 Init()函数

init 函数里面: mipi_set_cmd_mode()和 mipi_gen_write()的寄存器分别为:

#define R_DSI_HOST_CMD_MODE_CFG (0x68) // cmd mode lp/hs host TX 模式选择

#define R_DSI_HOST_GEN_HDR (0x6C) //DATA TYPE 等 mipi 数据包头等信息

1.1.6.27 CMD_MODE_CFG

0x00000068			Command mode configuration(0x00000000)								CMD_MODE_CFG					
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved							max_rd_pkt_size	Reserved				dcs_lw_tx	dcs_sr_0p_tx	dcs_sw_1p_tx	dcs_sw_0p_tx
Type	RO							RW	RO				RW	RW	RW	RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	peripheral_tuning_tx	gen_lw_tx	gen_sr_2p_tx	gen_sr_1p_tx	gen_sr_0p_tx	gen_sw_2p_tx	gen_sw_1p_tx	gen_sw_0p_tx	cmc_tx	exq_tx	pps_tx	peripheral_shutdown_tx	color_mode_off_tx	color_mode_on_tx	ack_req_st_en	tear_fx_en
Type	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Command mode configuration

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 25]	RO	NA	0	
max_rd_pkt_size	[24]	RW	NA	0	This bit configures the maximum read packet size command transmission type: 0: High-speed 1: Low-power
reserved	[23: 20]	RO	NA	0	
dcs_lw_tx	[19]	RW	NA	0	This bit configures the DCS long write packet command transmission type: 0: High-speed 1: Low-power
dcs_sr_0p_tx	[18]	RW	NA	0	This bit configures the DCS short read packet with zero parameter command transmission type: 0: High-speed 1: Low-power
dcs_sw_1p_tx	[17]	RW	NA	0	This bit configures the DCS short write packet with one parameter command transmission type: 0: High-speed 1: Low-power

dcsw_sw_0p_tx	[16]	RW	NA	0	This bit configures the DCS short write packet with zero parameter command transmission type: 0: High-speed 1: Low-power
peripheral turnon_tx	[15]	RW	NA	0	This bit configures the Peripheral Turn On packet transmission type: 0: High-speed 1: Low-power
gen_lw_tx	[14]	RW	NA	0	This bit configures the Generic long write packet command transmission type: 0: High-speed 1: Low-power
gen_sr_2p_tx	[13]	RW	NA	0	This bit configures the Generic short read packet with two parameters command transmission type: 0: High-speed 1: Low-power

gen_sr_1p_tx	[12]	RW	NA	0	This bit configures the Generic short read packet with one parameter command transmission type: 0: High-speed 1: Low-power
gen_sr_0p_tx	[11]	RW	NA	0	This bit configures the Generic short read packet with zero parameter command transmission type: 0: High-speed 1: Low-power
gen_sw_2p_tx	[10]	RW	NA	0	This bit configures the Generic short write packet with two parameters command transmission type: 0: High-speed 1: Low-power
gen_sw_1p_tx	[9]	RW	NA	0	This bit configures the Generic short write packet with one parameter command transmission type: 0: High-speed 1: Low-power
gen_sw_0p_tx	[8]	RW	NA	0	This bit configures the Generic short write packet with zero parameter command transmission type: 0: High-speed 1: Low-power
cmc_tx	[7]	RW	NA	0	This bit configures the Compression Mode Command packet transmission type: 0: High-speed 1: Low-power
exq_tx	[6]	RW	NA	0	This bit configures the Execute Queue Command packet transmission type:

					0: High-speed 1: Low-power
pps_tx	[5]	RW	NA	0	This bit configures the Picture Parameter Set packet transmission type: 0: High-speed 1: Low-power
peripheral shutdown_tx	[4]	RW	NA	0	This bit configures the Peripheral Shut Down packet transmission type: 0: High-speed 1: Low-power
color mode off_tx	[3]	RW	NA	0	This bit configures the Color Mode Off packet transmission type: 0: High-speed 1: Low-power
color mode on_tx	[2]	RW	NA	0	This bit configures the Color Mode On packet transmission type: 0: High-speed 1: Low-power
ack_rqst_en	[1]	RW	NA	0	When set to 1, this bit enables the acknowledge request after each packet transmission.
tear_fx_en	[0]	RW	NA	0	When set to 1, this bit enables the tearing effect acknowledge request.

1.1.6.28 GEN_HDR

0x0000006C		Generic packet header configuration(0x00000000)										GEN_HDR				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved										gen_wc_msbyte					
Type	RO										RW					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	gen_wc_lsbyte								gen_wc		gen_dt					
Type	RW								RW		RW					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Generic packet header configuration

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 24]	RO	NA	0	
gen_wc_msbyte	[23: 16]	RW	NA	0	This field configures the most significant byte of the header packet's word count for long packets or data 1 for short packets.
gen_wc_lsbyte	[15: 8]	RW	NA	0	This field configures the least significant byte of the header packet's Word count for long packets or data 0 for short packets.
gen_vc	[7: 6]	RW	NA	0	This field configures the virtual channel id of the header packet.
gen_dt	[5: 0]	RW	NA	0	This field configures the packet data type of the header packet.

2.2 mipi info

```
static struct info_mipi lcd_st7782m_mipi_info = {
    .work_mode = SPRDFB_MIPI_MODE_VIDEO,
    .video_bus_width = 24, /*18,16*/
    .lan_number = 1,
    .phy_freq = 300*1000,
    .h_sync_pol = SPRDFB_POLARITY_POS,
    .v_sync_pol = SPRDFB_POLARITY_POS,
    .de_pol = SPRDFB_POLARITY_POS,
    .te_pol = SPRDFB_POLARITY_POS,
    .color_mode_pol = SPRDFB_POLARITY_NEG,
    .shut_down_pol = SPRDFB_POLARITY_NEG,
    .timing = &lcd_st7782m_mipi_timing,
    .ops = NULL,
};
```

相关参数:work_mode

寄存器:

#define R_DSI_HOST_MODE_CFG (0x18) //video/cmd mode 配置

Work mode 在具体代码里还涉及到其他寄存器的配置，如；


```
/bsp/u-boot15_sprdroidq/drivers/video/sprdfb/sprdfb_dsi.c
```

```
int32_t sprdfb_dsi_ready(struct sprdfb_device *dev){};
```

1.1.6.7 DSI_MODE_CFG

0x00000018			Video or Command mode selection(0x00000000)									DSI_MODE_CFG				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved															cmd_video_mode

Type	RO																RW
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Video or Command mode selection

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 1]	RO	NA	0	
cmd_video_mode	[0]	RW	NA	0	This bit configures the operation mode: 0: Video mode 1: Command mode

相关参数: video_bus_width

寄存器:

```
#define R_DSI_HOST_DPI_COLOR_CODE (0x20) //Format 24bit
```

1.1.6.9 DPI_VIDEO_FORMAT

0x00000020			DPI video format(0x00000000)									DPI_VIDEO_FORMAT				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved									loosely18_en	dpi_video_mode_format					
Type	RO									RW	RW					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DPI video format

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 7]	RO	NA	0	
loosely18_en	[6]	RW	NA	0	When set to 1, this bit activates loosely packed variant to 18-bit configurations.
dpi video mode format	[5: 0]	RW	NA	0	This field configures the DPI color coding as follows: 0000: 16-bit configuration 1 0001: 16-bit configuration 2 0010: 16-bit configuration 3 0011: 18-bit configuration 1 0100: 18-bit configuration 2 0101: 24-bit 0110: 20-bit YCbCr 4:2:2 loosely packed 0111: 24-bit YCbCr 4:2:2 1000: 16-bit YCbCr 4:2:2 1001: 30-bit 1010: 36-bit 1011: 12-bit YCbCr 4:2:0 1100: Compression Display Stream 1101-1111: 12-bit YCbCr 4:2:0

相关参数: lan_number

寄存器:

```
#define R_DPHY_IF_CFG (0xA4)
```

1.1.6.42 LANE_NUM_CONFIG

0x000000A4			DPHY number of lanes configuration(0x00000003)									LANE_NUM_CONFIG				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved														lane_num	
Type	RO														RW	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1

DPHY number of lanes configuration

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 2]	RO	NA	0	
lane_num	[1: 0]	RW	NA	0x3	This field configures the number of active data lanes: 00: One data lane (lane 0) 01: Two data lanes (lanes 0 and 1) 10: Three data lanes (lanes 0, 1, and 2) 11: Four data lanes (lanes 0, 1, 2, and 3)

相关参数: phy_freq

源文件: /bsp/u-boot15_sprdroidq/drivers/video/sprdfb/dsi_1_21a/mipi_dsih_api.c

mipi_dsih_set_lp_clock()

```
{
    tx_escape_division = (instance->phy_freq>>4);
    tx_escape_division = (tx_escape_division+9999)/10000;//ceiling calc
    mipi_dsih_hal_tx_escape_division(instance, (uint8_t)tx_escape_division);
}
```

mipi_dsih_hal_tx_escape_division () 函数写入的相关寄存器:

#define R_DSI_HOST_CLK_MGR (0x5C)

1.1.6.24 TX_ESC_CLK_CONFIG

0x0000005C			Configuration of the clock dividers for tx_esc_clk(0x00000000)									TX_ESC_CLK_CONFIG				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	tx_esc_clk_config															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Configuration of the clock dividers for tx_esc_clk

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 16]	RO	NA	0	
tx_esc_clk_config	[15: 0]	RW	NA	0	This field indicates the division factor for the TX Escape clock source (lanebyteclk). The values 0 and 1 stop the TX_ESC clock generation.

在/u-boot15_sprddroidq/drivers/video/sprdfb/sprdfb_dsi.c 的 dsi_dpi_init() 函数，phy_feq 参与了其他变量的运算。

相关参数：

h_sync_pol:

v_sync_pol:

de_pol:

color_mode_pol:

shut_down_pol

寄存器：

```
#define R_DSI_HOST_DPI_CFG_POL (0x14)
```

此寄存器 DSI 已经没有，所以这些参数设置对 DSI 没有影响。

2.3 mipi_timing

```
static struct timing_rgb lcd_st7782m_mipi_timing = {  
    .hfp = 80, /* unit: pixel */  
    .hbp = 80,  
    .hsync = 20,  
    .vfp = 4, /*unit: line*/  
    .vbp = 4,  
    .vsync = 1,  
};
```

timing: 参考文档: SL8521E DSI 控制器介绍 1.0.pdf, 第二章的详细介绍, 需要经过公式换算。

寄存器:

```
#define R_DSI_HOST_VID_HSA_TIME      (0x28)  
#define R_DSI_HOST_VID_HBP_TIME      (0x28)  
#define R_DSI_HOST_VID_HLINE_TIME    (0x2C)  
#define R_DSI_HOST_VID_VSA_LINES     (0x30)  
#define R_DSI_HOST_VID_VBP_LINES     (0x30)  
#define R_DSI_HOST_VID_VFP_LINES     (0x30)  
#define R_DSI_HOST_VID_VACTIVE_LINES (0x34)
```

1.1.6.11 VIDEO_LINE_HBLK_TIME

0x00000028			Horizontal blanking time(0x00000000)									VIDEO_LINE_HBLK_TIME				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	video_line_hsa_time															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	video_line_hbp_time															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Horizontal blanking time

Field Name	Bit	Type	Set/Clear	Reset Value	Description
video line hsa time	[31: 16]	RW	NA	0	This field configures the Horizontal Synchronism Active period in lane byte clock cycles.
video line hbp time	[15: 0]	RW	NA	0	This field configures the Horizontal Back Porch period in lane byte clock cycles.

1.1.6.12 VIDEO_LINE_TIME

0x0000002C			Video line time(0x00000000)									VIDEO_LINE_TIME				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	video_line_time															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Video line time

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 16]	RO	NA	0	
video_line_time	[15: 0]	RW	NA	0	This field configures the size of the total line time (HSA+HBP+HACT+HFP) counted in lane byte clock cycles.

1.1.6.13 VIDEO_VBLK_LINES

0x00000030			Vertical blanking period lines(0x00000000)									VIDEO_VBLK_LINES				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved		vsa_lines									vbp_lines				
Type	RO		RW									RW				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	vbp_lines						vfp_lines									
Type	RW						RW									
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Vertical blanking period lines

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 30]	RO	NA	0	
vsa_lines	[29: 20]	RW	NA	0	This field configures the Vertical Synchronism Active period measured in number of horizontal lines.
vbp_lines	[19: 10]	RW	NA	0	This field configures the Vertical Back Porch period measured in number of horizontal lines.
vfp_lines	[9: 0]	RW	NA	0	This field configures the Vertical Front Porch period measured in number of horizontal lines.

1.1.6.14 VIDEO_VACTIVE_LINES

0x00000034			Vertical resolution(0x00000000)									VIDEO_VACTIVE_LINES				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	Reserved															
Type	RO															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	Reserved		vactive_lines													
Type	RO		RW													
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Vertical resolution

Field Name	Bit	Type	Set/Clear	Reset Value	Description
reserved	[31: 14]	RO	NA	0	
vactive_lines	[13: 0]	RW	NA	0	This field configures the Vertical Active period measured in number of horizontal lines.

2.4 panel spec struct

```

struct panel_spec lcd_st7782m_mipi_spec = {
    .width = 240,
    .height = 320,
    .fps = 60,
    .type = LCD_MODE_DSI,
    .direction = LCD_DIRECT_NORMAL,
    .info = {
        .mipi = &lcd_st7782m_mipi_info
    },
    .ops = &lcd_st7782m_mipi_operations,
};
  
```

相关参数: width 和 height

在 sprdfb_dsi.c 的 dsi_dpi_init() 函数中,

width => h_active_pixels, h_total_pixels,

h_active_pixels = panel->width;

h_total_pixels = panel->width + mipi->timing->hsync + mipi->timing->hbp + mipi->timing->hfp;

height => v_active_lines, v_total_lines,

v_active_lines = panel->height;

v_total_lines = panel->height + mipi->timing->vsync + mipi->timing->vbp + mipi->timing->vfp;

在 mipi_dsih_api.c 的 mipi_dsih_dpi_video() 函数里面,

h_active_pixels => video_size, 若是 burst mode, video_size 直接写到以下寄存器, 若是 non burst mode, 需要经过一些计算得出 video_size。


```
#define R_DSI_HOST_VID_PKT_SIZE (0x24)
```

1.1.6.10 VIDEO_PKT_CONFIG

0x00000024			video packet configuration(0x00000000)									VIDEO_PKT_CONFIG				
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	video_line_chunk_num															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	video_pkt_size															
Type	RW															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

video packet configuration

Field Name	Bit	Type	Set/Clear	Reset Value	Description
video_line_chunk_num	[31: 16]	RW	NA	0	This register configures the number of chunks to be transmitted during a Line period (a chunk consists of a video packet and a null packet). If set to 0 or 1, the video line is transmitted in a single packet. If set to 1, the packet is part of a chunk, so a null packet follows it if vid_null_size > 0. Otherwise, multiple chunks are used to transmit each video line.
video_pkt_size	[15: 0]	RW	NA	0	This field configures the number of pixels in a single video packet. For 18-bit not loosely packed data types, this number must be a multiple of 4. For YCbCr data types, it must be a multiple of 2, as described in the DSI specification.

h_total_pixels:

```
mipi_dsih_hal_dpi_hline(instance, (uint16_t)((video_params->h_total_pixels * ratio_clock_xPF) / PRECISION_FACTOR));
```

```
hs_timeout = ((video_params->h_total_pixels * video_params->v_active_lines) + (DSIH_PIXEL_TOLERANCE * bytes_per_pixel_x100) / 100);
```

```
for (counter = 0x80; (counter < hs_timeout) && (counter > 2); counter--)
```

```
{
```

```
    if ((hs_timeout % counter) == 0)
```



```

{
    mipi_dsih_hal_timeout_clock_division(instance, counter);
    mipi_dsih_hal_lp_rx_timeout(instance, (uint16_t)(hs_timeout / counter));
    mipi_dsih_hal_hs_tx_timeout(instance, (uint16_t)(hs_timeout / counter));
    break;
}
}

```

v_active_lines:

```

mipi_dsih_hal_dpi_vactive(instance, video_params->v_active_lines);
#define R_DSI_HOST_VID_VACTIVE_LINES      (0x34)

mipi_dsih_hal_dpi_vfp(instance, video_params->v_total_lines - (video_params->v_back_porch_lines
+ video_params->v_sync_lines + video_params->v_active_lines));
/* HS timeout */
hs_timeout = ((video_params->h_total_pixels * video_params->v_active_lines) + (DSIH_PIXEL_TOLERANCE * bytes_per_pixel_x100) / 100);
for (counter = 0x80; (counter < hs_timeout) && (counter > 2); counter--)
{
    if ((hs_timeout % counter) == 0)
    {
        mipi_dsih_hal_timeout_clock_division(instance, counter);
        mipi_dsih_hal_lp_rx_timeout(instance, (uint16_t)(hs_timeout / counter));
        mipi_dsih_hal_hs_tx_timeout(instance, (uint16_t)(hs_timeout / counter));
        break;
    }
}
}

```

v_total_lines:

```

mipi_dsih_hal_dpi_vfp(instance, video_params->v_total_lines - (video_params->v_back_porch_lines
+ video_params->v_sync_lines + video_params->v_active_lines));

```

相关参数: direction

暂时没有用到。